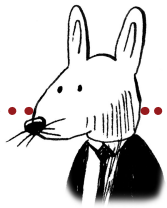
fundamental concepts: how do machines learn? 2



In this chapter

- What are deep learning, machine learning, and artificial intelligence?
- What are parametric models and nonparametric models?
- What are supervised learning and unsupervised learning?
- How can machines learn?

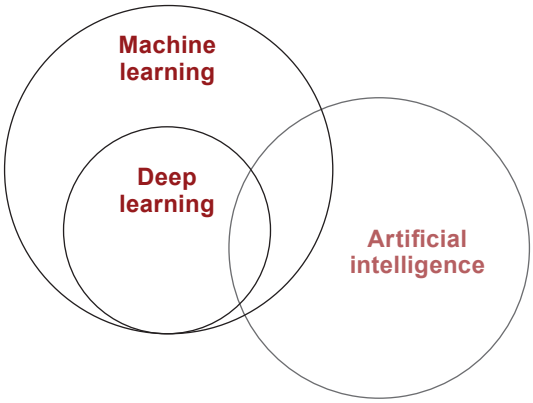
> "Machine learning will cause every successful IPO win in five years."
>
> —Eric Schmidt, Google executive chairman, keynote speech, Cloud Computing Platform conference, 2016

What is deep learning?

Deep learning is a subset of methods for machine learning.

Deep learning is a subset of machine learning, which is a field dedicated to the study and development of machines that can learn (sometimes with the goal of eventually attaining general artificial intelligence).

In industry, deep learning is used to solve practical tasks in a variety of fields such as computer vision (image), natural language processing (text), and automatic speech recognition (audio). In short, deep learning is a subset of *methods* in the machine learning toolbox, primarily using *artificial neural networks*, which are a class of algorithm loosely inspired by the human brain.



Notice in this figure that not all of deep learning is focused around pursuing generalized artificial intelligence (sentient machines as in the movies). Many applications of this technology are used to solve a wide variety of problems in industry. This book seeks to focus on teaching the fundamentals of deep learning behind both cutting-edge research and industry, helping to prepare you for either.

What is machine learning?

> A field of study that gives computers the ability to learn without being explicitly programmed.
>
> —Attributed to Arthur Samuel

Given that deep learning is a subset of machine learning, what is machine learning? Most generally, it is what its name implies. Machine learning is a subfield of computer science wherein *machines learn* to perform tasks for which they were *not explicitly programmed*. In short, machines observe a pattern and attempt to imitate it in some way that can be either direct or indirect.

Machine learning ~= Monkey see, monkey do

I mention direct and indirect imitation as a parallel to the two main types of machine learning: *supervised* and *unsupervised*. Supervised machine learning is the direct imitation of a pattern between two datasets. It's always attempting to take an input dataset and transform it into an output dataset. This can be an incredibly powerful and useful capability. Consider the following examples (*input* datasets in bold and *output* datasets in italic):

- Using the **pixels** of an image to detect the *presence* or *absence of a cat*
- Using the **movies you've liked** to predict more *movies you may like*
- Using someone's **words** to predict whether they're *happy* or *sad*
- Using weather sensor **data** to predict the *probability of rain*
- Using car engine **sensors** to predict the optimal tuning *settings*
- Using news **data** to predict tomorrow's stock *price*
- Using an input **number** to predict a *number* double its size
- Using a raw **audio file** to predict a *transcript* of the audio

These are all supervised machine learning tasks. In all cases, the machine learning algorithm is attempting to imitate the pattern between the two datasets in such a way that it can *use one dataset to predict the other*. For any of these examples, imagine if you had the power to predict the *output* dataset given only the **input** dataset. Such an ability would be profound.

Supervised machine learning

Supervised learning transforms datasets.

Supervised learning is a method for transforming one dataset into another. For example, if you had a dataset called Monday Stock Prices that recorded the price of every stock on every Monday for the past 10 years, and a second dataset called Tuesday Stock Prices recorded over the same time period, a supervised learning algorithm might try to use one to predict the other.

If you successfully trained the supervised machine learning algorithm on 10 years of Mondays and Tuesdays, then you could predict the stock price on any Tuesday in the future given the stock price on the immediately preceding Monday. I encourage you to stop and consider this for a moment.

Supervised machine learning is the bread and butter of applied artificial intelligence (also known as narrow AI). It's useful for taking *what you know* as input and quickly transforming it into *what you want to know*. This allows supervised machine learning algorithms to extend human intelligence and capabilities in a seemingly endless number of ways.

The majority of work using machine learning results in the training of a supervised classifier of some kind. Even unsupervised machine learning (which you'll learn more about in a moment) is typically done to aid in the development of an accurate supervised machine learning algorithm.

For the rest of this book, you'll be creating algorithms that can take input data that is observable, recordable, and, by extension, *knowable* and transform it into valuable output data that requires logical analysis. This is the power of supervised machine learning.

Unsupervised machine learning

Unsupervised learning groups your data.

Unsupervised learning shares a property in common with supervised learning: it transforms one dataset into another. But the dataset that it transforms into is *not previously known or understood*. Unlike supervised learning, there is no "right answer" that you're trying to get the model to duplicate. You just tell an unsupervised algorithm to "find patterns in this data and tell me about them."

For example, *clustering a dataset into groups* is a type of unsupervised learning. Clustering transforms a sequence of *datapoints* into a sequence of *cluster labels*. If it learns 10 clusters, it's common for these labels to be the numbers 1–10. Each datapoint will be assigned to a number based on which cluster it's in. Thus, the dataset turns from a bunch of datapoints into a bunch of labels. Why are the labels numbers? The algorithm doesn't tell you what the clusters are. How could it know? It just says, "Hey scientist! I found some structure. It looks like there are groups in your data. Here they are!"

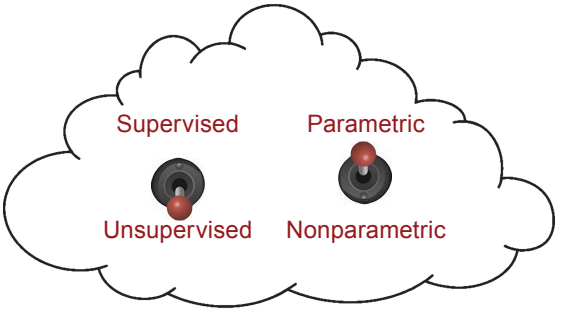
I have good news! This idea of clustering is something you can reliably hold onto in your mind as the definition of unsupervised learning. Even though there are many forms of unsupervised learning, *all forms of unsupervised learning can be viewed as a form of clustering*. You'll discover more on this later in the book.

Check out this example. Even though the algorithm didn't tell what the clusters are named, can you figure out how it clustered the words? (Answer: 1 == cute and 2 == delicious.) Later, we'll unpack how other forms of unsupervised learning are also just a form of clustering and why these clusters are useful for supervised learning.

Parametric vs. nonparametric learning

Oversimplified: Trial-and-error learning vs. counting and probability

The last two pages divided all machine learning algorithms into two groups: supervised and unsupervised. Now, we're going to discuss another way to divide the same machine learning algorithms into two groups: parametric and nonparametric. So, if we think about our little machine learning cloud, it has two settings:



As you can see, there are really four different types of algorithms to choose from. An algorithm is either unsupervised or supervised, and either parametric or nonparametric. Whereas the previous section on supervision is about the *type of pattern* being learned, parametricism is about the way the learning is *stored* and often, by extension, the *method for learning*. First, let's look at the formal definitions of parametricism versus nonparametricism. For the record, there's still some debate around the exact difference.

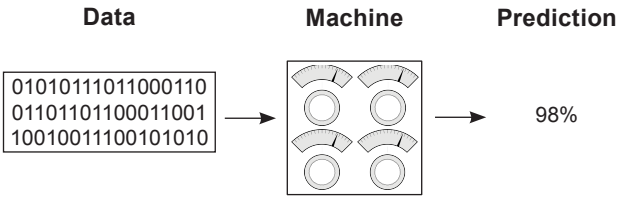
> A parametric model is characterized by having a fixed number of parameters, whereas a nonparametric model's number of parameters is *infinite* (determined by data).

As an example, let's say the problem is to fit a square peg into the correct (square) hole. Some humans (such as babies) just jam it into all the holes until it fits somewhere (parametric). A teenager, however, may count the number of sides (four) and then search for the hole with an equal number (nonparametric). Parametric models tend to use trial and error, whereas nonparametric models tend to count. Let's look closer.

Supervised parametric learning

Oversimplified: Trial-and-error learning using knobs

Supervised parametric learning machines are machines with a fixed number of knobs (that's the parametric part), wherein learning occurs by turning the knobs. Input data comes in, is processed based on the angle of the knobs, and is transformed into a *prediction*.



Learning is accomplished by turning the knobs to different angles. If you're trying to predict the probability that the Red Sox will win the World Series, then this model would first take data (such as sports stats like win/loss record or average number of toes per player) and make a prediction (such as 98% chance). Next, the model would observe whether or not the Red Sox actually won. After it knew whether they won, the learning algorithm would *update the knobs* to make a more accurate prediction the next time it sees the *same or similar input data*.

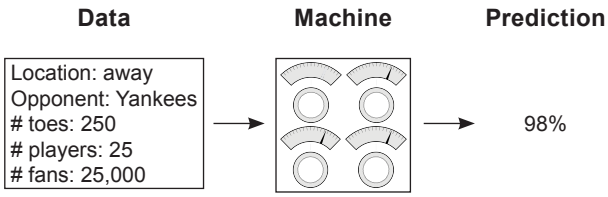
Perhaps it would "turn up" the "win/loss record" knob if the team's win/loss record was a good predictor. Inversely, it might "turn down" the "average number of toes" knob if that datapoint wasn't a good predictor. This is how parametric models learn!

Note that the entirety of what the model has learned can be captured in the positions of the knobs at any given time. You can also think of this type of learning model as a search algorithm. You're "searching" for the appropriate knob configuration by trying configurations, adjusting them, and retrying.

Note further that the notion of trial and error isn't the formal definition, but it's a common (with exceptions) property to parametric models. When there is an arbitrary (but fixed) number of knobs to turn, some level of searching is required to find the optimal configuration. This is in contrast to nonparametric learning, which is often count based and (more or less) adds new knobs when it finds something new to count. Let's break down supervised parametric learning into its three steps.

Step 1: Predict

To illustrate supervised parametric learning, let's continue with the sports analogy of trying to predict whether the Red Rox will win the World Series. The first step, as mentioned, is to gather sports statistics, send them through the machine, and make a prediction about the probability that the Red Sox will win.



Step 2: Compare to the truth pattern

The second step is to compare the prediction (98%) with the pattern you care about (whether the Red Sox won). Sadly, they lost, so the comparison is

Pred: 98% > **Truth**: 0%

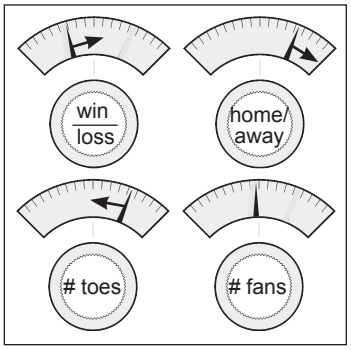
This step recognizes that if the model had predicted 0%, it would have perfectly predicted the upcoming loss of the team. You want the machine to be accurate, which leads to step 3.

Step 3: Learn the pattern

This step adjusts the knobs by studying both how *much* the model missed by (98%) and what the input data *was* (sports stats) at the time of prediction. This step then turns the knobs to make a more accurate prediction given the input data.

In theory, the next time this step saw the same sports stats, the prediction would be lower than 98%. Note that each knob represents the *prediction's sensitivity to different types of input data*. That's what you're changing when you "learn."

Adjusting sensitivity by turning knobs

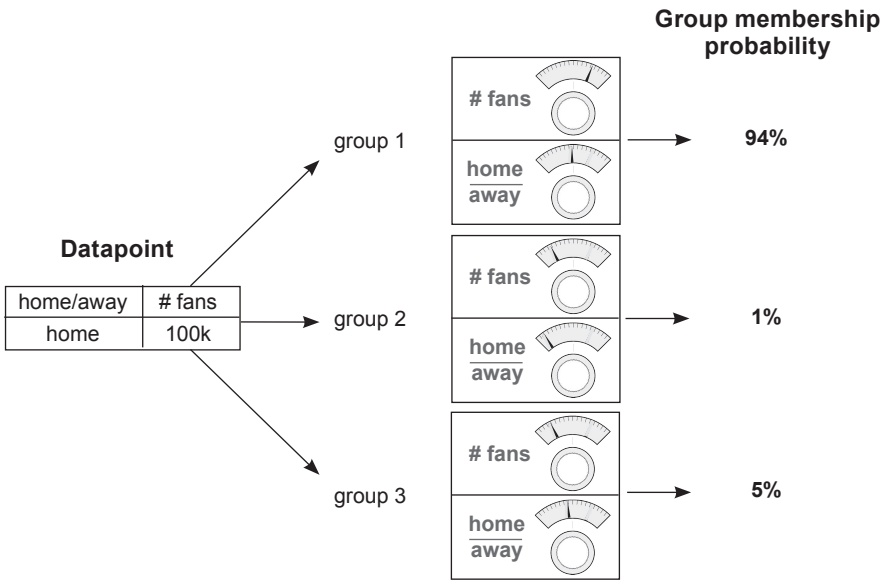


Unsupervised parametric learning

Unsupervised parametric learning uses a very similar approach. Let's walk through the steps at a high level. Remember that unsupervised learning is all about grouping data. Unsupervised *parametric* learning uses knobs to group data. But in this case, it usually has several knobs for each group, each of which maps the input data's affinity to that particular group (with exceptions and nuance—this is a high-level description). Let's look at an example that assumes you want to divide the data into three groups.

Home or away	# fans
home	**100k**
away	50k
home	**100k**
home	**99k**
away	50k
away	*10k*
away	*11k*

In the dataset, I've identified three clusters in the data that you might want the parametric model to find. They're indicated via formatting as **group 1**, group 2, and *group 3*. Let's propagate the first datapoint through a trained unsupervised model, as shown next. Notice that it maps most strongly to **group 1**.

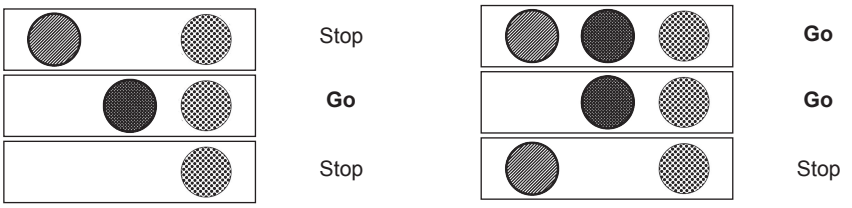


Each group's machine attempts to transform the input data to a number between 0 and 1, telling us the *probability that the input data is a member of that group*. There is a great deal of variety in how these models train and their resulting properties, but at a high level they adjust parameters to transform the input data into its subscribing group(s).

Nonparametric learning

Oversimplified: Counting-based methods

Nonparametric learning is a class of algorithm wherein the number of parameters is based on data (instead of predefined). This lends itself to methods that generally count in one way or another, thus increasing the number of parameters based on the number of items being counted within the data. In the supervised setting, for example, a nonparametric model might count the number of times a particular color of streetlight causes cars to "go." After counting only a few examples, this model would then be able to predict that *middle* lights always (100%) cause cars to go, and *right* lights only sometimes (50%) cause cars to go.



Notice that this model would have three parameters: three counts indicating the number of times each colored light turned on and cars would go (perhaps divided by the number of total observations). If there were five lights, there would be five counts (five parameters). What makes this simple model *nonparametric* is this trait wherein the number of parameters changes based on the data (in this case, the number of lights). This is in contrast to parametric models, which start with a set number of parameters and, more important, can have more or fewer parameters purely at the discretion of the scientist training the model (regardless of data).

A close eye might question this idea. The parametric model from before seemed to have a knob for each input datapoint. Most parametric models still have to have some sort of *input* based on the number of classes in the data. Thus you can see that there is a *gray area* between parametric and nonparametric algorithms. Even parametric algorithms are somewhat influenced by the number of classes in the data, even if they aren't explicitly counting patterns.

This also illuminates that *parameters* is a generic term, referring only to the set of numbers used to model a pattern (without any limitation on how those numbers are used). Counts are parameters. Weights are parameters. Normalized variants of counts or weights are parameters. Correlation coefficients can be parameters. The term refers to the set of numbers used to model a pattern. As it happens, deep learning is a class of parametric models. We won't discuss nonparametric models further in this book, but they're an interesting and powerful class of algorithm.

Summary

In this chapter, we've gone a level deeper into the various flavors of machine learning. You learned that a machine learning algorithm is either supervised or unsupervised and either parametric or nonparametric. Furthermore, we explored exactly what makes these four different groups of algorithms distinct. You learned that supervised machine learning is a class of algorithm where you learn to predict one dataset given another and that unsupervised learning generally groups a single dataset into various kinds of clusters. You learned that parametric algorithms have a fixed number of *parameters* and that nonparametric algorithms adjust their number of parameters based on the dataset.

Deep learning uses neural networks to perform both supervised and unsupervised prediction. Until now, we've stayed at a conceptual level as you got your bearings in the field as a whole and your place in it. In the next chapter, you'll build your first neural network, and all subsequent chapters will be *project based*. So, pull out your Jupyter notebook, and let's jump in!